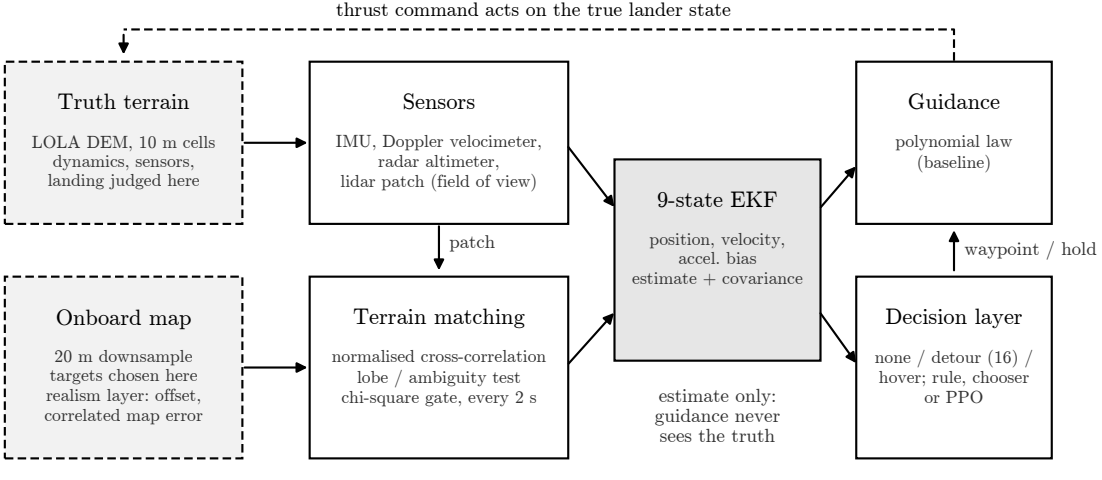
thrust command acts on the true lander state
Truth terrain
LOLA DEM, 10 m cells
dynamics, sensors,
landing judged here
Sensors
IMU, Doppler velocimeter,
radar altimeter,
lidar patch (field of view)
Guidance
polynomial law
(baseline)
patch
9-state EKF
position, velocity,
accel. bias
estimate + covariance
waypoint / hold
Onboard map
20 m downsample
targets chosen here
realism layer: offset,
correlated map error
Terrain matching
normalised cross-correlation
lobe / ambiguity test
chi-square gate, every 2 s
Decision layer
none / detour (16) /
hover; rule, chooser
or PPO
estimate only:
guidance never
sees the truth